WEEK 9 / LESSON 3 / 50 MINUTES

L3

Precision Farming for India

They cannot afford traditional precision agriculture. But a ₹6,000 phone + AI can deliver disease diagnosis, yield prediction, irrigation guidance. This is the highest-leverage AI work on the planet.

Does your family have farmland? What would your parents ask the AI?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

800 million Indians depend on farms averaging 1.08 hectares.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Why AgriTech is India's next frontier.
2. 4 flagship applications.
3. Satellite + CV for monitoring.
4. AI for smallholder farmers.
5. IoT + edge for on-field.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
4 flagship applications
Key concept of this lesson.

02
Satellite CV — free +...
District-level monitoring

03
Smallholder UX design
For ₹6,000 phones

04
IoT + edge on-field
LoRaWAN + Jetson

01

PART 1 OF 4

4 flagship applications

Key concept of this lesson.

CONCEPT 1 OF 4

4 flagship applications

A core idea you'll use throughout this lesson.

DEFINITION

4 flagship applications

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Satellite CV — free + powerful

Sentinel-2: 10 m res, 5-day revisit.
NDVI for crop health.
CNN segmentation for crop-type.
Time-series for anomaly detection.
Google Earth Engine — free.
STACK
Sentinel + GBM

Free. Available today.

KEY POINTS

- Sentinel-2: 10 m res, 5-day revisit
- NDVI for crop health
- CNN segmentation for crop-type
- Time-series for anomaly detection

CONCEPT 3 OF 4

Smallholder UX design

Feature-phone + voice first.
Bhashini regional-language.
Offline-first.
< 2 GB RAM target.
Actionable outputs — "water tomorrow".
PRINCIPLE
Actions, not charts

Farmers don't want dashboards.

DEFINITION

Smallholder UX design

For ₹6,000 phones

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

IoT + edge on-field

Soil-moisture + pH + temp sensors.
LoRaWAN: 5-15 km range, ultra-low power.
Edge gateway — Pi or Jetson Nano.
On-device ML decisions.
Cloud sync when connectivity returns.
FIELD
LoRaWAN + edge

Works where WiFi doesn't.

KEY POINTS

- Soil-moisture + pH + temp sensors
- LoRaWAN: 5-15 km range, ultra-low power
- Edge gateway — Pi or Jetson Nano
- On-device ML decisions

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.
2.
3.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*Where will you apply this in your next project?*
Q2	*What was the single biggest 'aha' from this lesson?*
Q3	*Which habit will you start using this week?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Average farm size:

A 1.08 ha

B 5 ha

C 20 ha

D 0.1 ha

ANSWER · QUESTION 1 OF 3

Average farm size:

CORRECT ANSWER

A

1.08 ha

WHY

Smallholder-dominant

KNOWLEDGE CHECK · QUESTION 2 OF 3

Sentinel-2 resolution:

A 100 m

B 10 m

C 1 km

D 1 m

ANSWER · QUESTION 2 OF 3

Sentinel-2 resolution:

CORRECT ANSWER

B

10 m

WHY

10 m res, 5-day revisit

KNOWLEDGE CHECK · QUESTION 3 OF 3

LoRaWAN advantage:

A Speed

B Long-range + low-power + no SIM

C High cost

D Urban only

ANSWER · QUESTION 3 OF 3

LoRaWAN advantage:

CORRECT ANSWER

B

Long-range + low-power + no SIM

WHY

Ideal for farms

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1. 4 flagship applications
2. Satellite CV — free +...
3. Smallholder UX design
4. IoT + edge on-field

APPLY IT → ASSIGNMENT

Where AI most matters.

"800 million farmers. ₹6,000 phones. Your highest-leverage work."

DELIVERABLES

- NEXT
- Lesson 4 — EdTech.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 9

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.